UDCT v7.13 | Preregistration

Refined Jeans Sign-Boundary and Hessian Diagnostics in Hydrus I

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Status: prospective numerical protocol, no new v7.13 galaxy or Hessian results reported. Original public version dated 13 October 2026; revised after initial publication to freeze previously deferred numerical rules. This revision must be published before confirmatory execution.

1. Provenance and scientific scope

The frozen v7.12 post-registration contains 363 fixed decisions: N=274, P=48, I=41. All 24 inherited sign-change brackets stopped on numerical margins. The full 24-case inventory, original bounds and inherited scientific source hashes are retained unchanged. The objective is to distinguish numerically supported sign brackets from unresolved cases, not to derive a new gravity law.

2. Locked baseline and independent resolution check

Four inherited deciding arms: default and refined grids at inner cutoffs 0.002 and 0.0005 r_h. Default: radial=360, polar=24, azimuth=36, disk=120x64x12. Refined: radial=720, polar=48, azimuth=72, disk=240x128x24.

New validation arm, prescribed BEFORE any v7.13 scientific run: radial=1440, polar=96, azimuth=144, disk=480x256x48 at BOTH inner cutoffs. Use identical inherited physical equations, quadrature conventions, profile, outer boundary, source data, and common-radius interpolation. This is independent by resolution, NOT by numerical algorithm. Failure to construct the arm is a failed-validation outcome; do not replace it post hoc.

For each trial coordinate, require finite common-grid normalized Jeans second-moment profiles for all six arms and finite native-grid minima. Define E as the maximum absolute pairwise difference at any common sampled radius among the six profiles; tau=max(1e-8,5 E). This is a conservative numerical envelope, not a proof of continuum convergence. Archive each arm, E, tau, native minimum, and gates.

3. Sign acceptance and deterministic stopping

Define F as the minimum of the six common-grid minima and six native-grid minima. A robust positive point requires F > tau. A robust negative point requires all six common-grid profiles to be below -tau at at least one SHARED sampled radius and F < -tau. Otherwise classify as unresolved numerical margin. A strict opposite-sign endpoint pair is necessary for a validated sign bracket; a P/N category change alone is insufficient.

Process B01 through B24 in fixed order. For each: evaluate original inherited bracket endpoints, stop immediately if nonfinite, failed inherited isotropic/field gate, or either endpoint is margin-limited. If robust signs do not oppose, report invalid/no robust bracket. Otherwise bisect at the arithmetic midpoint, preserving opposite-sign endpoints. Stop at first margin-limited midpoint, bracket width <=0.001 for beta or <=0.01 pc for ell, or 20 bisections. No monotonicity or uniqueness assumed; even a width-qualified interval is NOT an exact critical coordinate.

4. Hessian: explicit estimator and limits

The target Hessian is H_ij = d_i d_j Phi_total of the same 3-D gravitational model. A 1-D Jeans sign test and the external Galactic tidal tensor are not substitutes. If a differentiable 3-D total potential is available, compute centered diagonal second differences and centered four-corner mixed differences, symmetrize, using steps h/r_h = 0.02, 0.01, 0.005 at fixed registered sampling locations. For diagonal terms use [Phi(x+h e_i)-2 Phi(x)+Phi(x-h e_i)]/h^2; for mixed terms use the four-corner stencil divided by 4h^2.

Accept Hessian convergence only if all entries are finite, max|H(.01)-H(.005)| <= max(1e-10 GM/r_h^3, 0.05 max|H(.005)|), and differences decrease between successive halvings (or both differences lie below the same floor). Otherwise report not converged. Report trace, eigenvalues and Frobenius norm for converged cases only. All Hessian-versus-Jeans associations are exploratory; no universal Hessian threshold is claimed.

5. Pre-execution gate and reporting

All six-arm implementation and Hessian evaluation code must be self-tested, hash recorded, and publicly accessible before confirmatory calculations. The accompanying archive contains the inherited historical runner and a protocol audit, NOT a completed v7.13 validation runner or 3-D Hessian solver. The protocol is fixed; implementation failure is reported, not resolved by unregistered relaxation. Preserve all attempts, errors, partial outputs, environment, timestamps and source hashes. Report all 24 outcomes, including failures.

6. Transparent version history and interpretation

Original public Zenodo preregistration: 13 October 2026. This 14 October revision fixes previously deferred criteria. It must not be described as present in the original timestamped text. The researcher must confirm that no new v7.13 scientific simulation preceded this revision. P is only a positive sampled Jeans candidate; N and I are conditional on the stationary closure and numerical bounds. No equilibrium, positive distribution function, dynamical stability, observed fit, or general exclusion of MOND follows.

Appendix: inherited 24 target inventory

ID	Axis	Law	Fixed	Initial inherited bracket
B01	beta	BASE	1.0	0.35 to 0.4
B02	beta	BASE	1.2	0.4 to 0.45
B03	beta	BASE	1.5	0.45 to 0.5
B04	ell_pc	BASE	0.4	1.0 to 1.2
B05	ell_pc	BASE	0.45	1.2 to 1.5
B06	ell_pc	BASE	0.5	1.5 to 2.0
B07	beta	A-f2	1.0	0.4 to 0.45
B08	beta	A-f2	1.2	0.4 to 0.45
B09	beta	A-f2	1.5	0.45 to 0.475
B10	ell_pc	A-f2	0.45	1.2 to 1.5
B11	ell_pc	A-f2	0.5	1.5 to 2.0
B12	beta	A-Qs	1.0	0.35 to 0.4
B13	beta	A-Qs	1.2	0.35 to 0.4
B14	beta	A-Qs	1.5	0.4 to 0.45
B15	beta	A-Qs	2.0	0.475 to 0.5
B16	ell_pc	A-Qs	0.4	1.2 to 1.5
B17	ell_pc	A-Qs	0.45	1.5 to 1.75
B18	ell_pc	A-Qs	0.5	2.0 to 2.25
B19	beta	A-Qd	1.0	0.35 to 0.4
B20	beta	A-Qd	1.2	0.4 to 0.45
B21	beta	A-Qd	1.5	0.45 to 0.475
B22	ell_pc	A-Qd	0.4	1.0 to 1.2
B23	ell_pc	A-Qd	0.45	1.2 to 1.5
B24	ell_pc	A-Qd	0.5	1.75 to 2.0

References and archive

v7.12 postregistration: https://doi.org/10.5281/zenodo.23296444

v7.12 preregistration: https://doi.org/10.5281/zenodo.23280972

GitHub: https://github.com/wspaik5/UDCT_v7.13_Preregistration_Jeans_Boundary_Refinement

Machine-readable locked criteria: preregistration_plan.json; protocol self-check: UDCT_v7_13_Protocol_Audit.py; unchanged source hashes in JSON.